

Enterprise Architecture, Security & Operational Master Guide

Full-lifecycle research manuscript publishing, peer-review system, cloud infrastructure and operational deployment manual.

PUBLIC EDITION // ARCHITECTURAL SPECIFICATION // SECRETS REDACTED

Project Name	Shaoor (Academic Journal & Research Publication Engine)
Live Custom Domain	https://shaoor.org https://www.shaoor.org
Vercel Deployment	https://shaoor-nine.vercel.app (Project: ceo-h-s-ns-projects/shaoor)
GitHub Repository	https://github.com/CEO-H-S-N/Shaoor (Branch: main)
System Principal	Dr. Shoukat Tilwani (Owner / Master Administrator)
Document Version	v2.4.0 (Production Release — September 2026)
Security Classification	UNRESTRICTED (Public Architecture & Developer Manual)

PUBLIC STAKEHOLDER EDITION

This public edition documents the system architecture, component boundaries, database schemas, and workflows. All private credentials, passwords, and tokens have been replaced with [REDACTED] indicators. It is safe for academic review and faculty onboarding.

1. Executive Summary & Platform Overview

Shaoor (■■■■■ — *Consciousness & Enlightenment*) is a premier academic journal and research dissemination platform. Engineered to bridge rigorous peer-review standards with modern web performance, Shaoor provides authors, peer reviewers, and journal editors with a unified digital workspace.

The platform eliminates the latency and fragmentation of traditional academic publishing portals by combining:

- **Self-Service Author Portal:** Guided manuscript submission with metadata extraction, multi-author ORCID attribution, keyword indexing, and automated APA citation generation.
- **Scientific Figure Gallery:** Dedicated database and cloud asset management for embedded high-resolution figures, schematics, and captions linked to manuscripts.
- **Double-Blind Review Queue:** Role-based reviewer workspace supporting qualitative critiques, score-based assessment, structured decisions (Accept, Reject, Request Revisions), and direct author feedback notifications.
- **Dynamic Editorial CMS:** Zero-downtime content management allowing the Owner to directly edit the Journal Scope, Author Guidelines, and Editorial Team directory from the web UI.
- **Automated Auth & Verification:** Multi-factor onboarding featuring secure email OTP codes, self-service password reset, and Google One-Click OAuth authentication.
- **AI Research Assistant:** Integrated Google Gemini Generative AI sidebar assisting researchers with scholarly literature synthesis and paper exploration.

2. System Architecture & Technology Stack

Layer	Technology	Specification & Role
Frontend Framework	Next.js 15 (App Router)	React 19, Server Components, TypeScript, CSS Modules (Zero Tailwind bloat)
Application Hosting	Vercel Edge Network	Serverless Node.js 20 runtime, Edge Middleware for JWT auth validation
DNS & Domain	Vercel Anycast DNS	Custom domain shaoor.org & www.shaoor.org via ns1/ns2.vercel-dns.com
Database Service	AWS RDS PostgreSQL 16	Managed db.t3.micro in eu-north-1 (Stockholm), SSL encrypted, automated backups
ORM & Schema	Prisma ORM v6	Type-safe database client, connection pooling, automated migrations
Object Storage	AWS S3	Private bucket in eu-north-1 with temporary IAM pre-signed upload/download URLs
Authentication	Auth.js (NextAuth v5)	Dual-strategy: Google OAuth 2.0 + Credentials with Bcrypt (cost factor 12)
Transactional Mail	SMTP / Nodemailer	Six-digit cryptographic OTP generation with 10-minute automated expiry
AI Assistant	Google Gemini API	Gemini 1.5 Flash streaming completions for contextual research inquiries

3. Role-Based Access Control (RBAC) & Governance

Shaoor enforces strict server-side authorization boundaries across four distinctive user classes. Permissions are validated on both Edge Middleware (route guard) and within Server Actions (database mutation guard).

Role	Privilege Scope	Administrative Capabilities & UI Visibility
MASTER OWNER (Dr. Shoukat Tilwani)	Root System Governance	- Provision new trusted Administrator accounts with zero developer intervention. - Direct in-place editing of About Us, Author Guidelines, and Team Directory. - Review, Accept, Revise, or Reject any manuscript with author feedback. - Ban / Deactivate accounts; override any reviewer decision.
ADMINISTRATOR	Editorial Review Queue	- Access the global paper review queue (/review). - Read submitted manuscripts, download PDFs, view attached scientific figures. - Submit formal review decisions (Accept, Reject, Request Revisions) with feedback. - View submission metrics. <i>Cannot edit CMS pages or create new Admins.</i>
CUSTOMER / AUTHOR	Research Submitter	- Submit new research papers with abstract, category, keywords, and figures. - Track real-time manuscript status (Submitted, Under Review, Revise, Published). - View reviewer feedback notes on rejected/revised papers and re-submit revisions. - Access APA citation generator and public research repository.
DESIGNER	Interface Auditor	Audit layout, typography, and styling across all responsive viewport states.

4. Database Schema & Data Models (Prisma ORM)

The underlying PostgreSQL database contains 12 normalized models designed for referential integrity, performance indexing, and audit logging:

Model	Primary Keys & Indexing	Key Attributes & Relational Logic
User	id (cuid), email (unique)	name, passwordHash, role, isActive, bannedReason, emailVerified. Relations to Papers, Reviews, AuditLogs.
Paper	id (cuid), authorId, status	title, abstract, status (DRAFT -> PUBLISHED), fileId, decisionMessage, decisionBy. Has many Figures and Versions.
PaperFigure	id (cuid), paperId	figureNumber, title, caption, fileId (AWS S3), fileName, fileSize. Cascades on Paper deletion.
PaperVersion	id (cuid), paperId + versionNumber	Maintains archival history of manuscript iterations across peer-review cycles.
Review	unique(paperId, reviewerId)	decision (ACCEPT, REJECT, REVISE), comments, privateNotes, score, isComplete.

EmailOtp	id (cuid), email index	otp, expiresAt (10m TTL), used (boolean). Prevents replay attacks for signups and password resets.
PageContent	id (cuid), slug (unique)	Dynamic CMS content for /about and /guidelines. Editable by Master Owner.
TeamMember	id (cuid), sort_order	Editorial board members: name, designation, institution, summary, image, email.
AuditLog	id (cuid), userId, action	OWASP compliance: tracks user registrations, authentication events, and administrative decisions.

5. Infrastructure Secrets & Credentials Directory

PUBLIC VARIABLE DIRECTORY

SANITIZED ENVIRONMENT TEMPLATE: Secrets below have been redacted for public safety. Operators should inject these variables into their local .env.local or Vercel Environment Variables dashboard.

Category / Service	Key / Parameter	Value / Host / Credentials
Master Account (Journal Owner)	Login Email Master Password	Email: [REDACTED_FOR_SECURITY] Password: [REDACTED_MASTER_KEY]
Staging Accounts (Dev Testing)	Admin / User Passwords	Dev Admin: [REDACTED_FOR_SECURITY] [REDACTED] Dev Author: [REDACTED_FOR_SECURITY] [REDACTED]
AWS RDS PostgreSQL (eu-north-1)	DATABASE_URL	Host: database-1.cxa2gagc85ck.eu-north-1.rds.amazonaws.com:5432 DB / User: shaoor / postgres Password: [REDACTED_DB_PASSWORD] Full Connection String: postgresql://[USER]:[PASSWORD]@[ENDPOINT]:5432/shaoor?schema=public
AWS S3 Asset Bucket (Manuscripts & Figures)	AWS_ACCESS_KEY_ID AWS_SECRET_ACCESS_KEY AWS_S3_BUCKET_NAME	Bucket: shaoor-papers-065768154566-eu-north-1-an Region: eu-north-1 Access Key ID: [REDACTED_ACCESS_KEY] Secret Access Key: [REDACTED_SECRET_KEY]
Google Cloud Console (OAuth Sign-In)	GOOGLE_CLIENT_ID GOOGLE_CLIENT_SECRET	Client ID: [GOOGLE_CLIENT_ID] Client Secret: [REDACTED_CLIENT_SECRET]
Google AI Studio (Gemini Assistant)	GOOGLE_GENERATIVE_AI_API_KEY	[REDACTED_GEMINI_API_KEY]
NextAuth / Auth.js (Session Encryption)	AUTH_SECRET	[32_CHAR_RANDOM_SECRET_KEY]
Domain & Nameservers (Namecheap -> Vercel)	DNS Delegation	Primary Nameserver: ns1.vercel-dns.com Secondary Nameserver: ns2.vercel-dns.com Apex & WWW: shaoor.org www.shaoor.org

6. Operational Playbook & Maintenance Runbook

A. How to Provision a New Academic Administrator Account

1. Sign in to <https://shaoor.org/login> using the Master Account (shouket.tilwani@gmail.com).
2. In the left administrative navigation bar, click **Make Admin Account**.
3. Enter the Professor or Editor's Full Name, Institutional Email, and assign a temporary password.
4. Click **Create Admin Account**. The system automatically registers the user with ADMIN role privileges.
5. The recipient can immediately log in and update their academic bio, ORCID, and password in their Profile Settings.

B. How to Edit Dynamic Journal Content (About Us, Guidelines, Team)

1. While signed in as the Master Owner, navigate to **/about**, **/author-guidelines**, or **/team**.
2. Click the **Edit Page** button displayed in the side panel or page banner.
3. Modify the title, mission statement, publication criteria, or add/remove editorial board members with photos and affiliations.
4. Click **Save Changes**. Content is persisted directly to AWS RDS PostgreSQL and reflected immediately worldwide.

C. How to Review, Approve, or Reject a Manuscript

1. Open **/review** from the top navigation or admin sidebar.
2. Click on any paper currently in SUBMITTED or UNDER_REVIEW status.
3. Examine the manuscript PDF in the integrated viewer and inspect attached figures.
4. In the decision module at the bottom: select **Approve**, **Reject**, or **Request Revision**.
5. Provide constructive peer-review notes. When submitted, the author receives an automated notification in their portal with your comments and an option to re-submit.

D. Routine Engineering & Database Maintenance

- **Schema Evolution:** Modify `prisma/schema.prisma`, then execute `npx prisma db push` or `npx prisma migrate dev`.
- **Vercel Production Deployments:** Pushing commits to the main branch of <https://github.com/CEO-H-S-N/Shaoor> automatically triggers an atomic production build and instant edge deployment.
- **Local Environment Testing:** Run `npm run dev` inside `frontend/` to launch the hot-reloading Next.js dev server on `http://localhost:3000`.

AUDIT & COMPLIANCE

SYSTEM AUDIT LOGGING: All administrative actions, paper reviews, user registrations, and account deactivations are automatically committed to the `audit_logs` table with client IP, user agent, and timestamp. Query this table in RDS for security reviews.